

Improving SIRENs for Image Representation: Learning Rate Scheduling, Shortcut Connections, and Self-Attention

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1. Introduction

Implicit Neural Representation (INR) is a deep learning technique that learns a function to return the value at a given coordinate or position in the data. One of the main advantages of INR is their ability to handle infinite resolution, making them suitable for high-dimensional signals or applications that require significant memory. INR is particularly useful in signal processing tasks such as images and videos. By compressing information about the data within the neural network, INR can reduce the data size and reconstruct it with high quality. However, early INR architectures confronted problem capturing high-frequency details effectively. For instance, ReLU, being piecewise linear with zero second derivatives, was unable to accurately represent high-frequency details. To address this issue, Vincent Sitzmann et al. proposed Sinusoidal Representation Networks (SIRENs) in their 2020 paper, "Implicit Neural Representations with Periodic Activation Functions." [3] SIRENs are particularly effective at representing complex signals and derivatives, excelling in capturing high-frequency details. They have demonstrated excellent performance across various domains, including images, videos, sounds, and boundary value problems. However, this report will focus specifically on their application to images.

In this report, we propose several methods to enhance the performance of SIRENs. The pixel coordinates of the images used in our experiments are 2D, so the SIRENs model has two input features, and since the images are grayscale, the model has one output channel. The baseline model is based on the publicly available code from the paper "Implicit Neural Representations with Periodic Activation Functions." [2] The specific structure of baseline SIRENs model matches a Figure 1. By modifying the baseline model, we discovered three tricks to improve the performance of SIRENs. The first trick involves Learning Rate Scheduling. The second trick is adding Shortcut Layers, similar to ResNet. The final trick is incorporating self-attention mechanisms within the SIREN model.

The performance of the models was evaluated using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM). PSNR measures the ratio between the maximum possible power of a signal and the power of corrupt-

method	PSNR (dB)	SSIM
baseline	35.840	0.923

Table 1. Baseline peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM)

ing noise, indicating the quality of the reconstructed image. SSIM assesses the similarity between two images by comparing their luminance, contrast, and structure, providing a perceptual measure of image quality. The baseline SIRENs model achieved a peak signal-to-noise ratio of 35.840 dB and a structural similarity index of 0.923 (Table 1). Figure 2 shows the image fitted using the baseline SIRENs model and the ground truth image. All three proposed tricks resulted in higher PSNR and SSIM values compared to the baseline model. In the following sections, each trick will be introduced in detail.

2. Experiments

2.1. Learning Rate Scheduling

Learning rate scheduling involves adjusting the learning rate during training to enhance model performance and convergence. Generally, a higher learning rate is used at the beginning and a lower learning rate towards the end. Initially, a larger learning rate helps the model explore the parameter space more broadly, enabling it to escape local minima and learn faster. As training progresses, reducing the learning rate allows the model to fine-tune the parameters more precisely, leading to better convergence and reduced oscillations around the optimal solution. A gradual decrease in the learning rate helps to stabilize the training process. To improve the performance of the SIRENs model, I employed cosine annealing learning rate scheduling.

I experimented with two methods of learning rate scheduling. The first method involved the commonly used approach of gradually decreasing the learning rate during training. The second method, conversely, involved gradually increasing the learning rate. For the gradually decreasing learning rate, the initial rate was set to 0.0001, and the final rate was set to 0.00001. For the gradually increasing learning rate, the initial rate was set to 0.0001, and the final

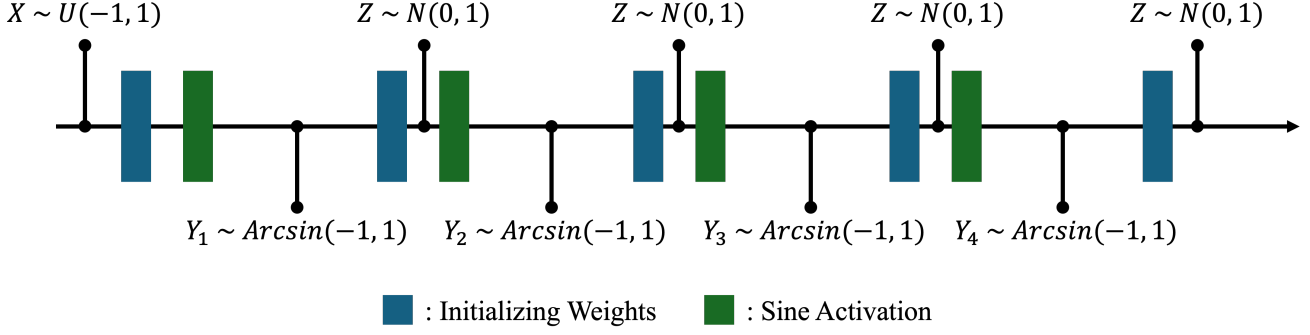


Figure 1. Baseline SIRENs Structure

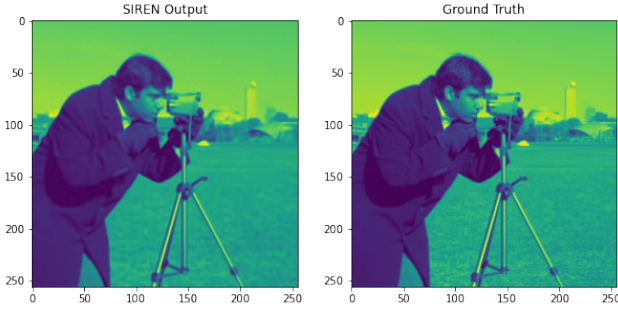


Figure 2. Image fitted using the baseline SIRENs model and the ground truth image

initial learning rate	final learning rate	PSNR (dB)	SSIM
0.0001	0.00001	33.207	0.895
0.0001	0.001	44.708	0.987

Table 2. Peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) when the learning rate was gradually decreased and when it was gradually increased

rate was set to 0.001. Contrary to expectations, the PSNR and SSIM values were higher when the learning rate was gradually increased compared to both the baseline and the gradually decreasing learning rate. Specifically, the PSNR reached 44.708 dB and the SSIM was 0.987 with the gradually increasing learning rate, which are significantly higher results (Table 2). On the other hand, the gradually decreasing learning rate resulted in a PSNR of 33.207 dB and an SSIM of 0.895, which are lower than the baseline (Table 2). In conclusion, for the SIRENs model, gradually increasing the learning rate during training yields better performance. Figure 3 shows the image fitted using the gradually decreasing learning rate SIRENs model and gradually increasing learning rate SIRENs model.

2.2. Adding Shortcut Layer

The introduction of shortcut layers in ResNet revolutionized the field of deep learning by demonstrating that adding

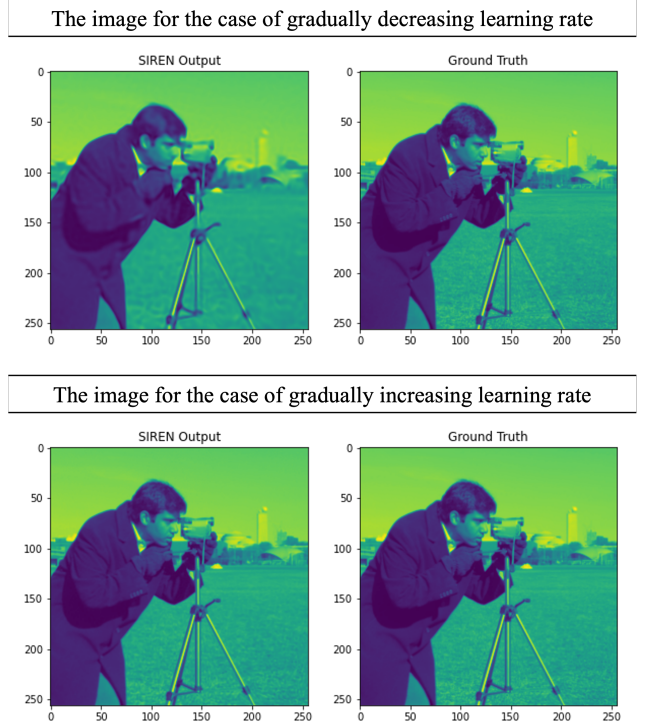


Figure 3. Image fitted using the gradually decreasing learning rate SIRENs model and gradually increasing learning rate SIRENs model

method	PSNR (dB)	SSIM
baseline	35.840	0.923
baseline + shortcut layers	39.171	0.962

Table 3. Peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) when the learning rate was gradually decreased and when it was gradually increased

these layers can significantly enhance model performance. Shortcut layers, also known as skip connections, allow the model to bypass one or more layers by adding the input of a layer directly to the output of a subsequent layer. These

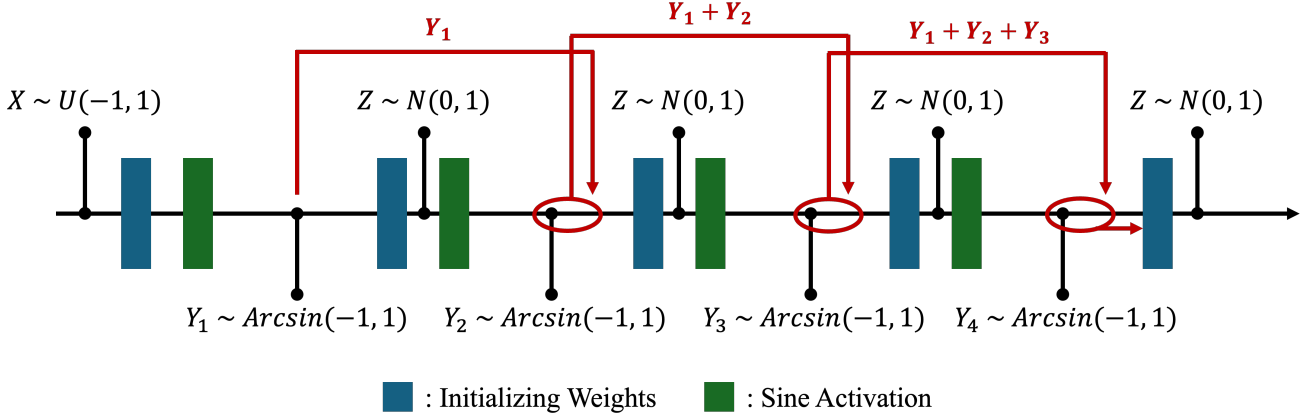


Figure 4. Adding shortcut layers to the SIRENs model

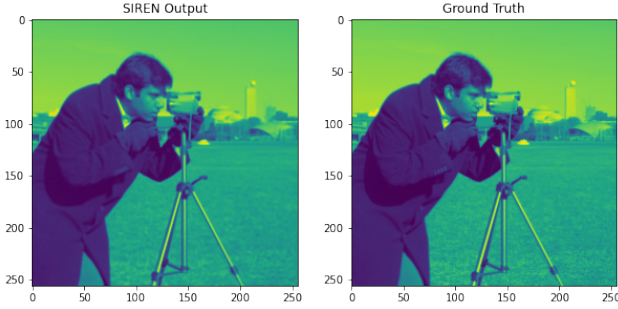


Figure 5. Image fitted using the SIRENs model with added shortcut layers

layers address the vanishing gradient problem, facilitating the training of much deeper networks without degradation. By preserving important features learned in earlier layers, shortcut layers lead to better convergence and improved accuracy. The success of ResNet has inspired the incorporation of shortcut layers into various other architectures. Consequently, we attempted to enhance the performance of the SIRENs model by adding shortcut layers.

As shown in Figure 4, shortcut layers were added to the SIRENs model. As a result, the PSNR of the fitted image improved to 39.171 dB, and the SSIM increased to 0.962, both higher than the baseline (Table 3). Just as in other deep learning models, adding shortcut layers to SIRENs can be expected to yield better performance. Figure 5 shows the image fitted using the SIRENs model with added shortcut layers.

2.3. Self-Attention

Initially popularized by the transformer model in the paper "Attention is All You Need" by Vaswani et al. in 2017[4, 1], self-attention allows models to weigh the importance of different parts of the input sequence, enabling more effective context understanding. This mechanism has

led to state-of-the-art (SOTA) performance in various natural language processing (NLP) tasks. Following its success in NLP, self-attention has also been applied to the computer vision domain, contributing to significant advancements in image classification. Therefore, this report suggests that self-attention could also enhance the performance of SIRENs when applied appropriately. By allowing the model to focus on relationships within the coordinates, self-attention could improve the representation capabilities of SIRENs.

There are five potential positions within the SIRENs structure where self-attention can be added (Figure 6). The results of adding self-attention at each of these positions and fitting the model to images are shown in Table 4. Adding self-attention immediately after the first SIRENs layer (Position 1) and after the last SIRENs layer (Position 5) resulted in significantly lower PSNR and SSIM values compared to other positions. Specifically, Position 1 yielded a PSNR of 29.548 dB and an SSIM of 0.814. Position 5 yielded a PSNR of 26.365 dB and an SSIM of 0.654. The best performance was observed when self-attention was added at Position 3, with a PSNR of 37.996 dB and an SSIM of 0.955. Position 2 and position 4 showed moderate performance. Position 2 achieved a PSNR of 35.342 dB and an SSIM of 0.923. Position 4 showed a PSNR of 36.075 dB and an SSIM of 0.904. These results suggest that incorporating self-attention in the middle of the SIRENs model is most effective, yielding better performance than the baseline. Applying self-attention at other positions results in performance similar to or worse than the baseline.

3. Conclusion

In this report, I explored various methods to enhance the performance of Sinusoidal Representation Networks (SIRENs) by experimenting with learning rate scheduling, adding shortcut layers, and incorporating self-attention

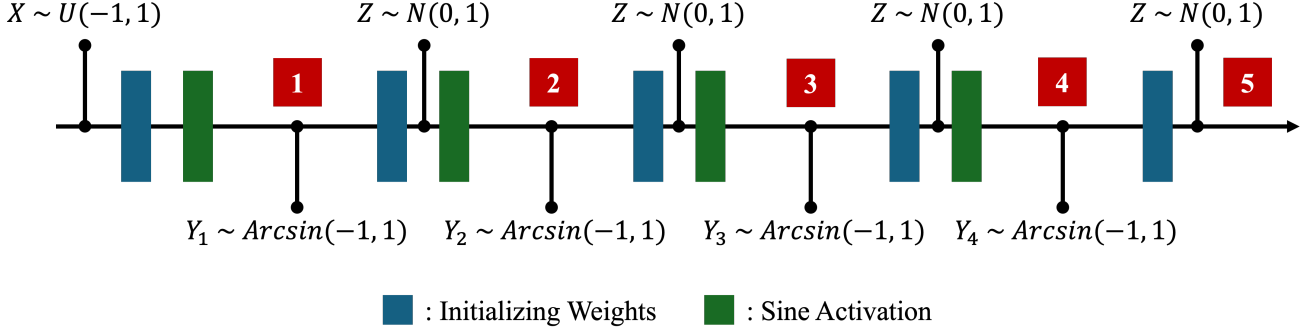


Figure 6. Five potential positions within the SIRENs structure where self-attention can be added

position	PSNR (dB)	SSIM
1	29.548	0.814
2	35.342	0.923
3	37.996	0.955
4	36.075	0.904
5	26.365	0.654

Table 4. Peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) for each position where self-attention is added

mechanisms.

Firstly, I investigated the impact of learning rate scheduling on model performance. The experiment revealed that a gradually increasing learning rate significantly improved the peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) compared to the baseline and a gradually decreasing learning rate. This finding suggests that for SIRENs, a progressively increasing learning rate during fitting image yields better results.

Secondly, I examined the effect of adding shortcut layers to the SIRENs model, inspired by the success of ResNet. Adding shortcut layers improved both PSNR and SSIM, indicating that preserving important features learned in earlier layers can significantly enhance model performance.

Lastly, I explored the inclusion of self-attention mechanisms within the SIRENs structure. By evaluating the performance of self-attention at various positions within the network, we found that adding self-attention in the middle of the SIRENs model (specifically Position 3) produced the best results. This suggests that self-attention can effectively enhance the representation capabilities of SIRENs when applied in the middle of the model, leading to superior image fitting performance.

In conclusion, our modifications to the baseline SIRENs model—adjusting the learning rate schedule, adding shortcut layers, and adding self-attention—resulted in improved performance.

References

- [1] Minqu Kanq. Transformer-pytorch, 2020. Accessed: 2024-06-16. [3](#)
- [2] Vincent Sitzmann. Implicit neural representations with periodic activation functions. GitHub, 2020. Accessed: 2024-06-01. [1](#)
- [3] Vincent Sitzmann, Julien Martel, Alexander Bergman, David Lindell, and Gordon Wetzstein. Implicit neural representations with periodic activation functions. *Advances in neural information processing systems*, 33:7462–7473, 2020. [1](#)
- [4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017. [3](#)